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Matematik NF

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Harmonic Analysis, Autumn 2026

MATP32

Exercises 3

1. Let $T, (b_Q)_{Q \in \mathcal{Q}}$ be defined as in the proof of Theorem 4 (Calderon-Zygmund decomposition) of the Lecture Notes. Prove that

$$\left| \left(T \sum_{Q \in \mathcal{Q}} b_Q \right) (x) \right| \leq \sum_{Q \in \mathcal{Q}} |(Tb_Q)(x)| \text{ for a.e. } x \in \mathbb{R}^d.$$

2. Let $\mathcal{Q}$ be a family of dyadic cubes. For $Q \in \mathcal{Q}$, let $Q^* = 2d^{1/2}Q$, and let c_Q denote the center of Q . Show that

$$\begin{aligned} \sup_{Q \in \mathcal{Q}, y \in Q} \int_{\mathbb{R}^d \setminus Q^*} |K(x-y) - K(x-c_Q)| dx \\ \leq \sup_{y \in \mathbb{R}^d} \int_{|x-c_Q| > 2|y-c_Q|} |K(x-y) - K(x-c_Q)| dx \\ = \sup_{y \in \mathbb{R}^d} \int_{|x| > 2|y|} |K(x-y) - K(x)| dx. \end{aligned}$$

3. Let δ_j, η be functions in $\mathcal{S}(\mathbb{R}^d)$ as defined the the lecture. Show that for any multi-indices $a, b \in \mathbb{N}_0^d$ and $|a| = M$, there exists a constant $C_{d,a,b,\eta}$ such that

$$|\partial^b (P_a \cdot \delta_j)(\xi)| \leq C_{d,a,b,\eta} 2^{j(|a|-M)} \quad (\xi \in \mathbb{R}^d),$$

where $P_a(\xi) = (i2\pi\xi)^a$ (note the multi-index notation!).

4. Let $K \in \mathcal{C}^\infty(\mathbb{R}^d \setminus \{0\})$ and suppose that for each multi-index $a \in \mathbb{N}_0^d$, there exists $C_a > 0$ with

$$|\partial^a K(x)| \leq C_a |x|^{-(|a|+d)} \quad (x \in \mathbb{R}^d \setminus \{0\}).$$

Show that K satisfies the Hörmander condition, i.e.

$$\sup_{y \in \mathbb{R}^d} \int_{|x| > 2|y|} |K(x) - K(x-y)| dx < \infty.$$

Var god vänd!

5. Show that $(X, \|\cdot\|_X)$ is a Banach space, where

$$X = \left\{ f \in L^1(\mathbb{R}^d) : f = \sum_{k=1}^{\infty} \mu - k a_{Q_k}, \text{ where } (\mu_k) \in \ell^1(\mathbb{N}), a_{Q_k} \text{ is an atom on } Q_k \text{ for each } k \in \mathbb{N} \right\}$$

and

$$\|f\|_X = \inf \left\{ \sum_{k=1}^{\infty} |\mu_k| : f = \sum_{k=1}^{\infty} \mu - k a_{Q_k}, \text{ where } (\mu_k) \in \ell^1(\mathbb{N}), a_{Q_k} \text{ is an atom on } Q_k \text{ for each } k \in \mathbb{N} \right\}.$$

The questions will be discussed in the seminar on Thursday, October 1.